

DekantPM

Mathematical Analysis

L²-Norm AMM for Continuous Prediction Markets
Bin Discretization • Distribution Trading • Solvency

Solana • Anchor 0.32 • u128 Fixed-Point Arithmetic

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1. Introduction: Continuous Prediction Markets

1.1 What Are Prediction Markets?

A prediction market is a speculative market where participants trade contracts whose payoffs depend on the outcome of future events. In a binary prediction market, a contract pays \$1 if event E occurs and \$0 otherwise. The market price reflects the crowd's aggregate belief about the probability of E.

1.2 From Discrete to Continuous

Traditional prediction markets handle discrete outcomes. But many real-world quantities are continuous: temperatures, prices, dates. A continuous prediction market elicits a full probability distribution over a continuous outcome space — the crowd's aggregate PDF over all possible answers.

1.3 The Goal

1. Aggregate beliefs — Allow participants to express nuanced probabilistic views.
2. Maintain a consensus distribution — The market state encodes the crowd's aggregate PDF.
3. Incentivize accuracy — Traders who move the distribution toward truth profit.
4. Provide liquidity — An AMM ensures trades can always execute without counterparties.

2. Mathematical Framework: The L₂-Norm AMM

2.1 Core Setup

Consider a market with N outcomes (or bins). We define:

- k (= total_minted): Total collateral deposited as complete sets.
- $h = (h_1, \dots, h_n)$: AMM reserves per outcome.
- $x = (x_1, \dots, x_n)$: Positions held by traders, where $x_i = k - h_i$.

2.2 The Invariant

The AMM enforces a constant L₂-norm on positions:

$$\|x\|_2 = \sqrt{\sum x_i^2} = k \quad \Leftrightarrow \quad \sum x_i^2 = k^2$$

This defines a hypersphere of radius k in position space. All valid AMM states lie on this surface.

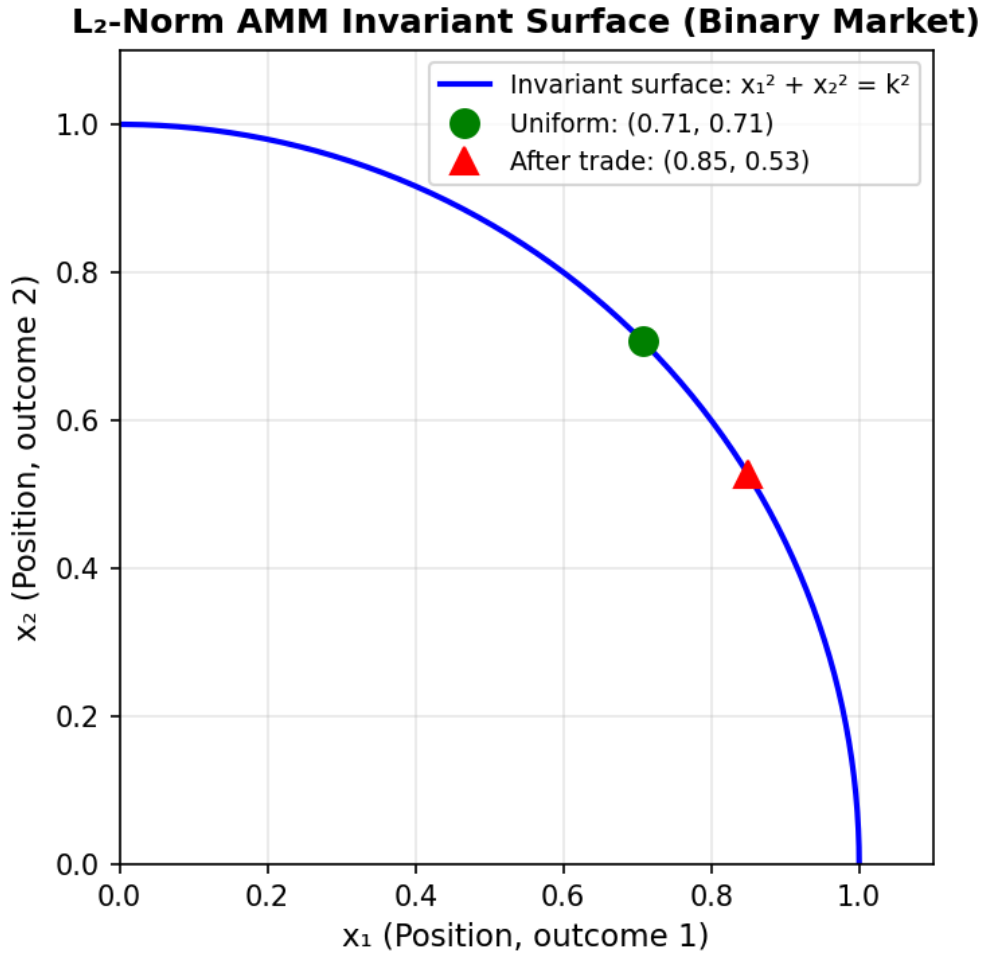


Figure 1: L_2 -norm invariant surface for a binary market.

2.3 Why L_2 -Norm? — Market Scoring Rule

Suppose the true distribution is $p = (p_1, \dots, p_n)$. A rational trader maximizes expected value on the invariant surface:

$$\max p \cdot x \quad \text{s.t.} \quad \|x\|_2 = k$$

By Cauchy-Schwarz, the optimum is:

$$x^* = k \cdot p / \|p\|_2$$

So at equilibrium $x_i \propto p_i$ — positions directly encode the probability distribution.

2.4 Probability Extraction

Linear extraction (true equilibrium probabilities):

$$p_i = x_i / \sum x_j$$

Quadratic extraction (used by DekantPM):

$$\hat{p}_i = x_i^2 / k^2$$

The quadratic values sum to 1 by the invariant ($\sum x_i^2 = k^2$). At equilibrium: $\hat{p}_i = p_i^2 / \|p\|_2^2$. The implementation scales by $S = 10^9$:

$$\hat{p}_{i_scaled} = x_i^2 \cdot S / k^2$$

2.5 Quadratic Pricing: Distortion Analysis

The quadratic formula is NOT the true probability — it is the squared and renormalized probability. This compresses values toward the extremes:

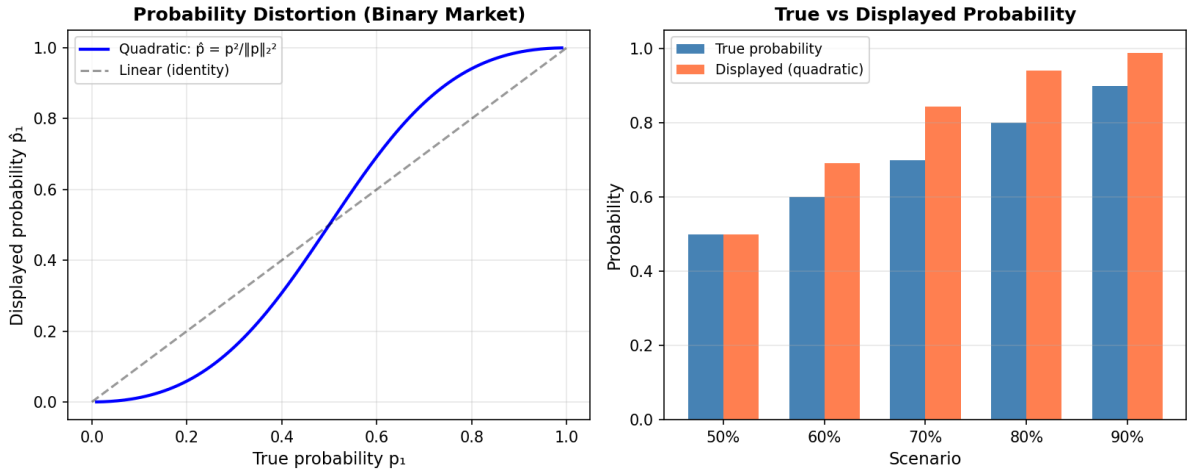


Figure 2: Quadratic probability distortion. True 70% displays as 84.5% in a binary market.

To recover true probabilities: $p_i = \sqrt{\hat{p}_i} / \sum \sqrt{\hat{p}_i}$. The quadratic formula is used because: (1) it sums to 1 by the invariant, (2) all AMM operations use it consistently, (3) outcome ordering is preserved.

Throughout this document, $\hat{p}_i$ refers to the quadratic extraction x_i^2/k^2 .

3. Market Initialization

With initial liquidity L and N outcomes:

$$k = L, \quad x_i = \lfloor \sqrt{(L^2/N)} \rfloor, \quad h_i = L - x_i, \quad k^2 = L^2$$

Initial probability: $\hat{p}_i = x_i^2/k^2 \approx 1/N$ (uniform).

Example (binary, $L = 1,000,000$):

$$x_1 = x_2 = 707,106 \quad h_1 = h_2 = 292,894 \quad \hat{p} = 0.5$$

4. Discrete Trading

4.1 Buy (Single Outcome)

Trader spends c collateral (after fees) to buy tokens of outcome i :

1. Mint complete sets: $h_j \leftarrow h_j + c$, $k' = k + c$.
2. Positions unchanged after mint: $x'_j = x_j$ for all j .
3. Solve for new position of target outcome:

$$x'_{i-1} = \sqrt{(k'^2 - \sum_{\{j \neq i\}} x_j^2)}$$

4. Tokens out = $x'_i - x_i$. Update $h_i = k' - x'_i$.

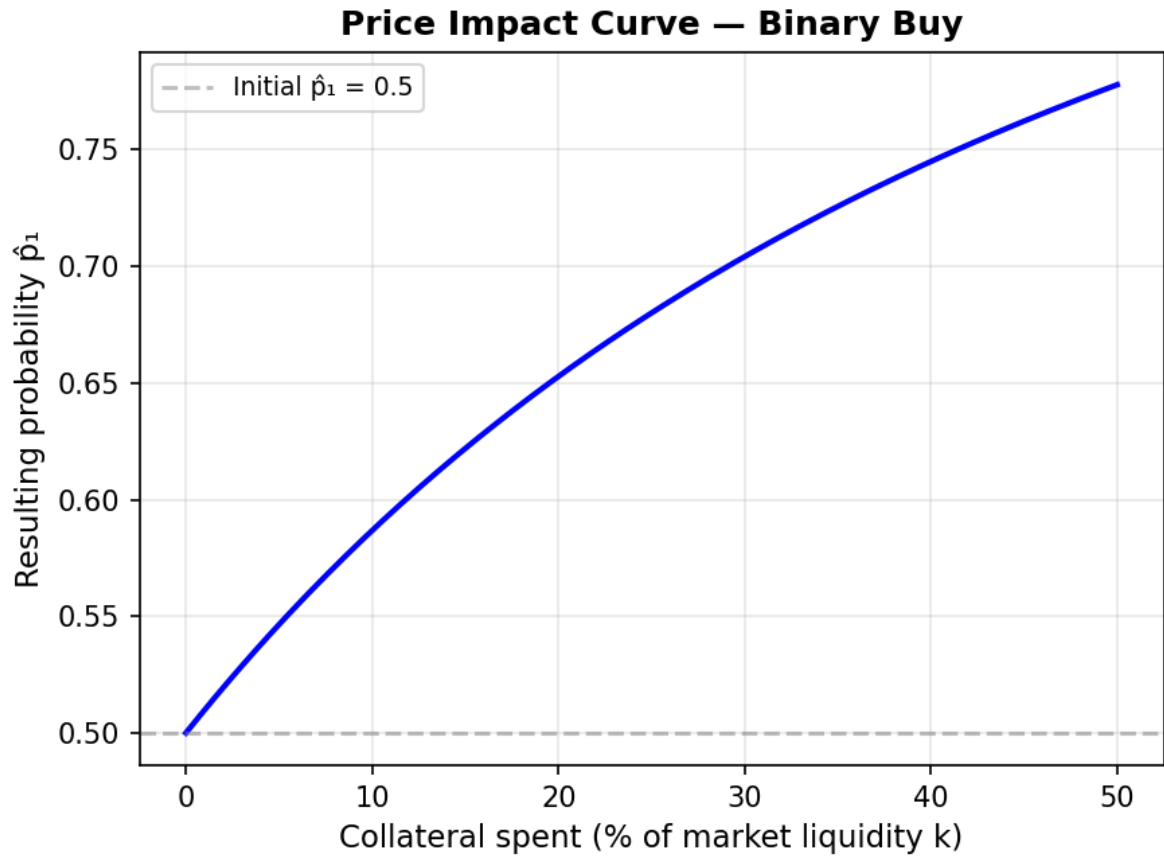


Figure 3: Price impact curve — concave relationship provides natural slippage protection.

4.2 Sell (Single Outcome)

Trader returns t tokens of outcome i :

1. Return: $h_i \leftarrow h_i + t$, so $x'_i = x_i - t$.
2. New total: $k' = \sqrt{(\sum x'_j)^2}$. Collateral out = $k - k'$.
3. Burn complete sets: $h_j \leftarrow h_j - (k - k')$ for all j .

5. Distribution Trading (Continuous Markets)

5.1 Bin Discretization

DekantPM discretizes $[a, b]$ into N bins. Bin j has center:

$$\text{center}_j = a + (2j + 1)(b - a) / (2N)$$

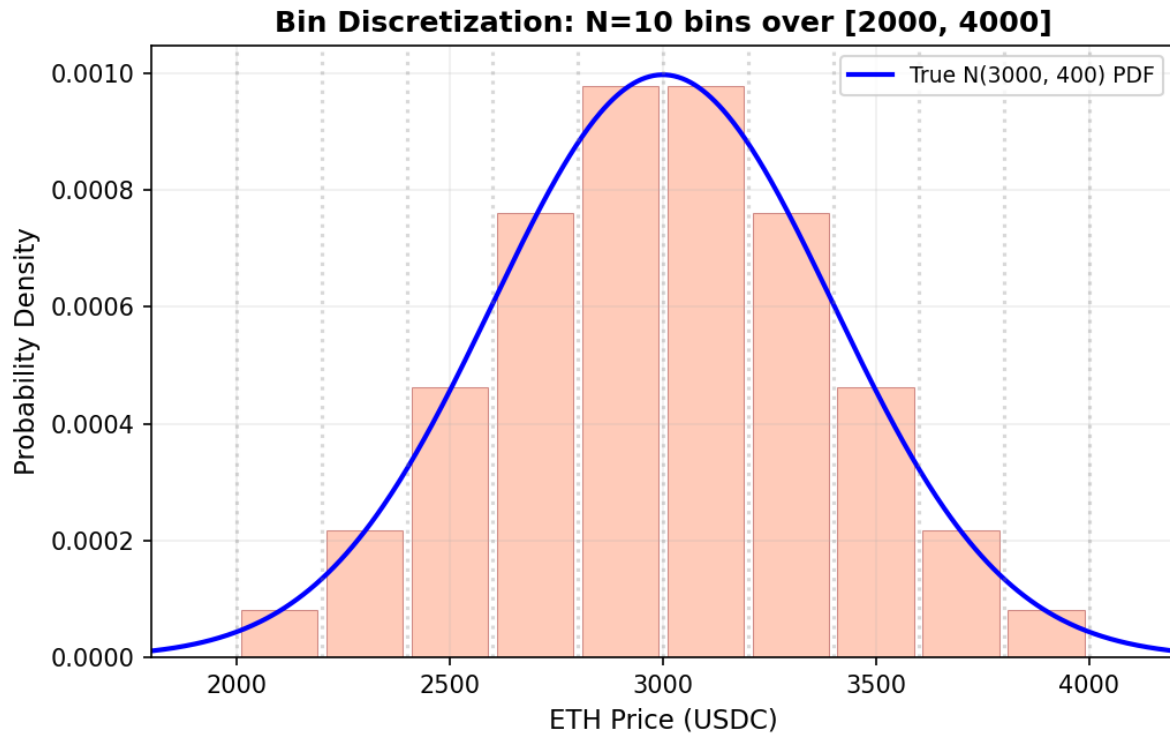


Figure 4: Bin discretization approximates the continuous Normal PDF as a histogram.

5.2 Normal PDF Bin Weights

For each bin j : $z_j = (\text{center}_j - \mu)/\sigma$. If $|z_j| > 5$, weight = 0. Otherwise:

$$w_j = \exp(-z_j^2/2) \quad \text{approximated by degree-4 Taylor polynomial}$$

$$\exp(-t/2) \approx 1 - t/2 + t^2/8 - t^3/48 + t^4/384 \quad (t = z^2)$$

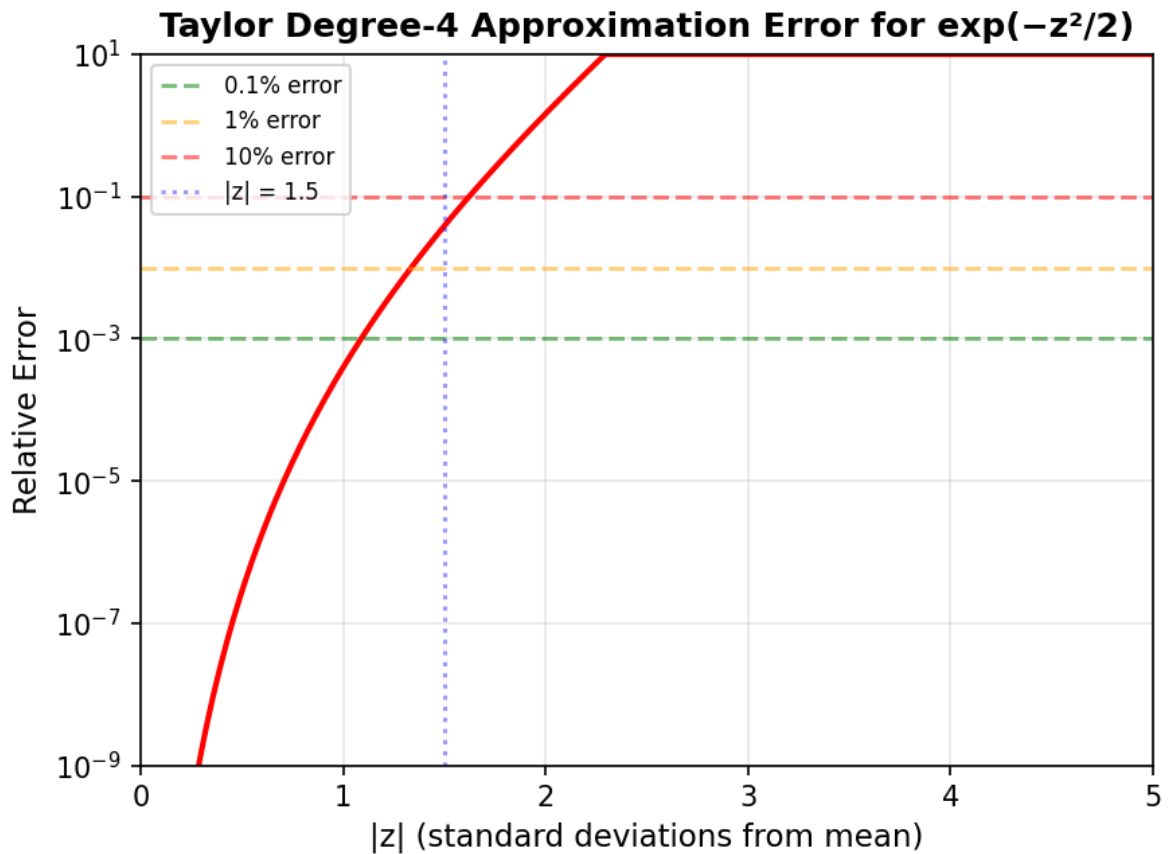


Figure 5: Taylor approximation error — excellent for $|z| < 1.5$, acceptable to $|z| \approx 3$.

Weights normalized: $W_j = w_j \cdot S / \sum w_m$, ensuring $\sum W_j = S = 10^9$.

5.3 Distribution Buy

Trader spends c to buy weighted bundle. Compute $XW = \sum x_j W_j$ and $W^2 = \sum W_j^2$, then:

$$\lambda = \sqrt{(XW^2 + W^2 \cdot \text{excess})} - XW \quad (\text{excess} = k'^2 - k^2)$$

$$\text{tokens_out}_j = \lambda \cdot W_j / W^2$$

5.4 Distribution Sell

Return T tokens proportional to W . Compute $k' = \sqrt{(\sum x_j'^2)}$, collateral out = $k - k'$.

5.5 Buy-to-Price

Target $\hat{p}_i = p^*/S$ for outcome i :

$$\text{collateral} = \sqrt{(\sum_{-i} \cdot S / (S - p^*))} - k$$

6. Liquidity Provision

6.1 Adding Liquidity

LP deposits d . Reserves scale proportionally:

$$h'_j = h_j \cdot (k + d) / k$$

Probabilities preserved. First LP: shares = L . Subsequent: shares = $L_{\text{total}} \cdot d / k$.

6.2 Removing Liquidity

Active: collateral_out = $k \cdot s / L_{\text{total}}$. Resolved: collateral_out = $h_{\text{winning}} \cdot s / L_{\text{total}}$. Plus fee share in both cases.

6.3 LP Economics

LPs earn fees (0.15%/trade default) and profit from mean reversion. They face divergence loss and adverse selection from informed traders.

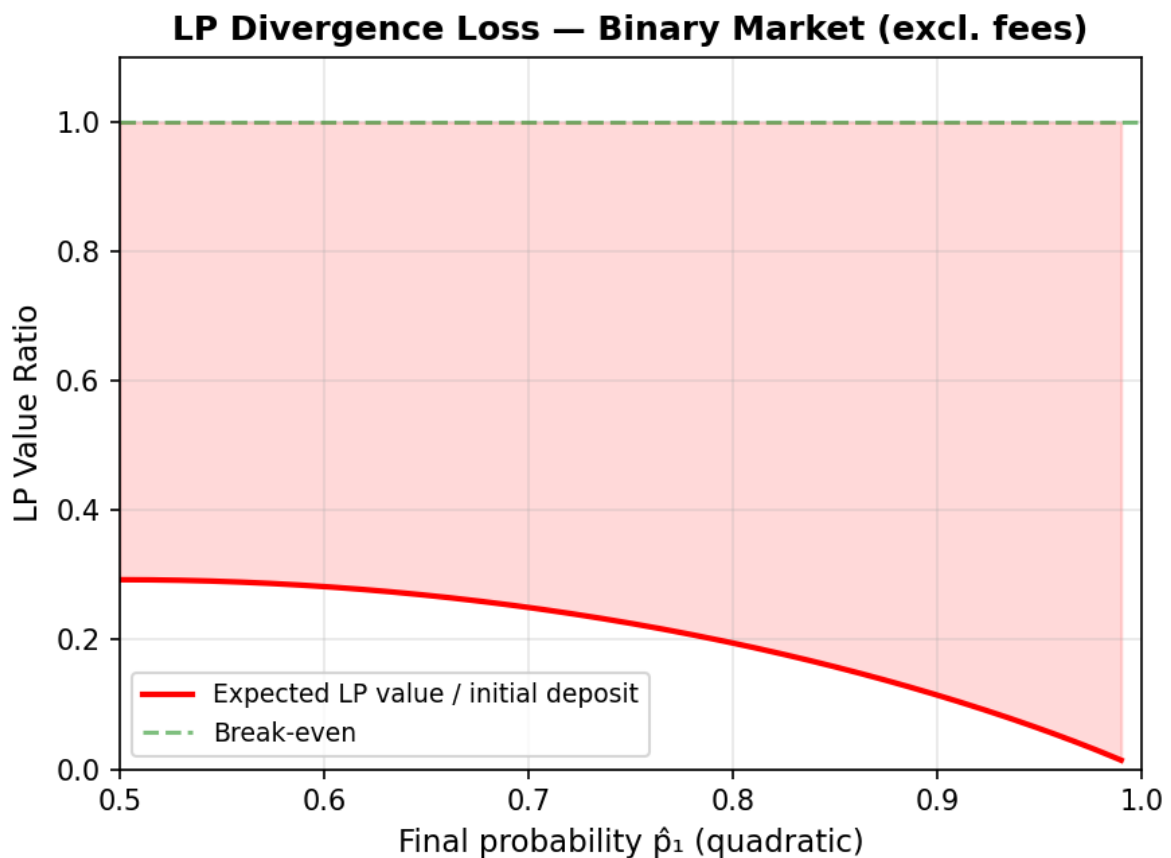


Figure 6: LP divergence loss in a binary market (excluding fees).

7. Resolution and Payout

7.1 Discrete Markets

Oracle provides winning outcome. Winners redeem 1:1 minus redemption fee (default 0.5%).

7.2 Continuous Markets

Oracle provides value $v \in [a, b]$. Winning bin = $\lfloor (v-a) \cdot N / (b-a) \rfloor$, clamped to $[0, N-1]$.

7.3 Solvency Guarantee

The vault always holds $\geq k$ collateral. Maximum redemption = $\max(x_i) \leq k$ (since $x_i^2 \leq \sum x_j^2 = k^2$). The system is always solvent. ■

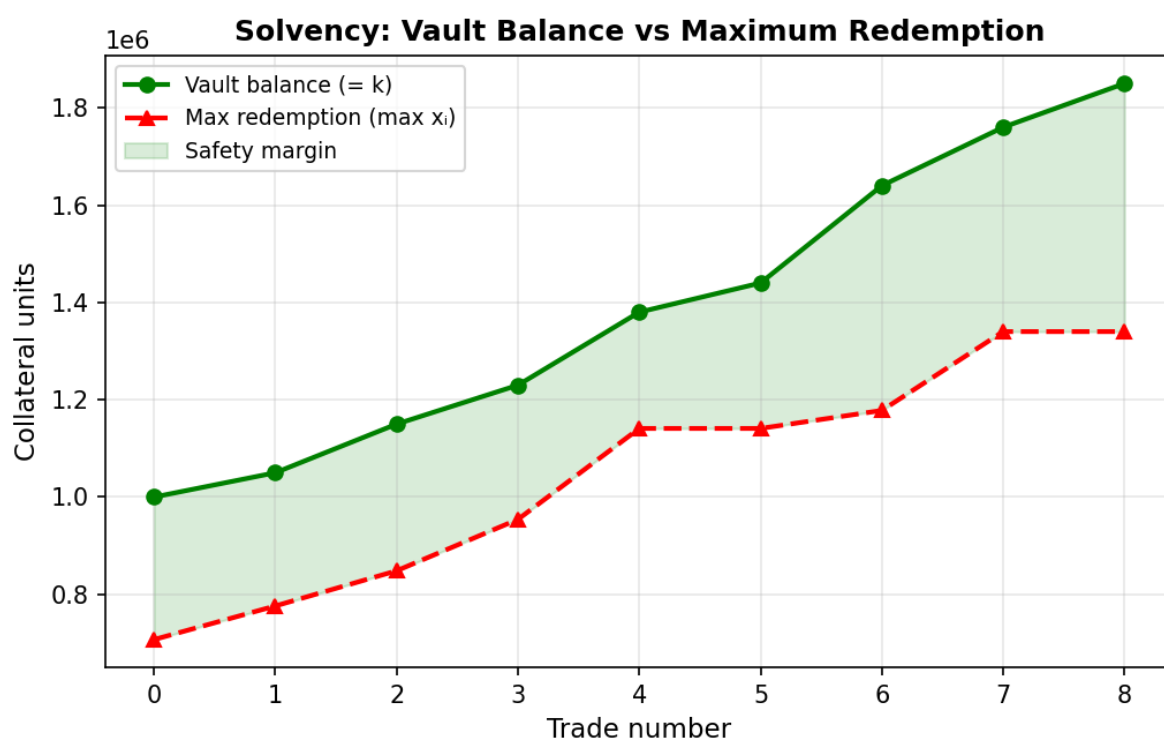


Figure 7: Solvency — vault balance always exceeds maximum possible redemption.

8. Worked Example: Continuous Market Lifecycle

8.1 Market Creation

Market: "ETH price March 31." Range [2000, 4000], N=5 bins, L = 1,000,000.

Bin	Range	Center
0	[2000,2400)	2200
1	[2400,2800)	2600
2	[2800,3200)	3000

3	[3200,3600)	3400
4	[3600,4000]	3800

$$k = 1,000,000 \quad x_i = 447,213 \quad h_i = 552,787 \quad p^*_i = 0.200$$

8.2 Trade 1 — Alice Buys Bin 2

Alice buys bin 2 with 100,000 collateral. Fee = 300, net = 99,700.

$$k' = 1,099,700 \quad \Sigma_{-2} = 4 \times 447,213^2 = 799,997,549,476$$

$$x'_2 = \sqrt{(1,209,340,090,000 - 799,997,549,476)} = 639,798$$

$$\text{tokens_out} = 192,585 \quad p^*_2 = 0.339 \quad (\text{was } 0.200)$$

8.3 Trade 2 — Bob's Distribution Buy N(3000, 400)

Bob spends 200,000 expressing $N(\mu=3000, \sigma=400)$. Bin weights concentrate on bins 1-3:

Bin	z	$\exp(-z^2/2)$	Weight ($\times 10^9$)
0	-2.0	0.135	54.5M
1	-1.0	0.607	244.3M
2	0.0	1.000	402.5M
3	+1.0	0.607	244.3M
4	+2.0	0.135	54.3M

After trade: bell-shaped distribution, bin 2 at ~40%.

8.4 Liquidity + Resolution

Carol adds 500,000 LP — depth increases, probabilities unchanged. Oracle reports ETH = \$3,150. Bin = $\lfloor (3150-2000) \times 5/2000 \rfloor = 2$. Alice's 192,585 tokens → net 191,623 (91.6% profit). Bob profits on bin-2 tokens only. Carol gets LP residual + fees.

Continuous Market Lifecycle — Probability Evolution

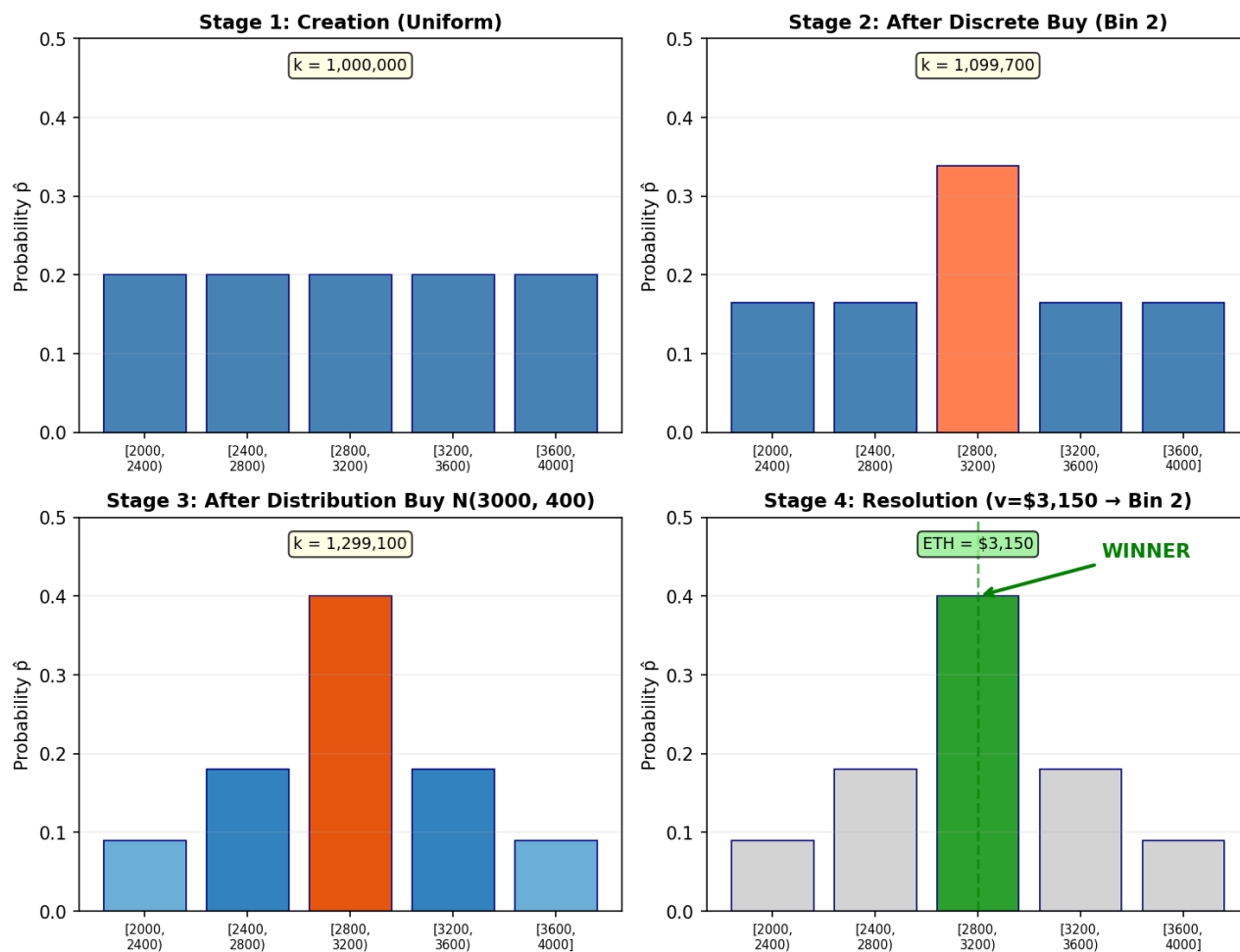


Figure 8: Probability evolution through the market lifecycle.

9. Fee Structure

```
total_fee = [G · trade_fee_bps / 10000]    (default 30 bps = 0.3%)
lp_fee = [total_fee · lp_share_bps / 10000] (default 50%)
protocol_fee = total_fee - lp_fee
redemption_fee = [payout · redemption_bps / 10000] (default 50 bps)
```

10. The Discretization Approach

10.1 Advantages

1. Simplicity — Same AMM engine for discrete and continuous markets.
2. On-chain efficiency — $O(N)$ operations, max $N=256$.
3. Composability — Individual bin tokens can be traded separately.
4. Exact solvency — Complete-set minting guarantees solvency.
5. No backing constraint — Unlike Paradigm's $\max(f) \leq b$.

10.2 Disadvantages

1. Resolution granularity — Same-bin values treated identically.
2. Approximation error — PDF becomes step function.
3. Fixed bins — Equal spacing may waste resolution.
4. Taylor error — Degrades for $|z| > 1.5$.
5. Normal-only — Cannot express skew or multimodality.

11. Limitations and Alternative Approaches

11.1 Single-Bin Resolution

Problem: Bin boundary creates payout discontinuity. Alternative: Pro-rata resolution distributes payouts to nearby bins proportional to distance from resolved value.

11.2 Fixed Number of Bins

Problem: N fixed at creation; consensus narrowing wastes bins. Alternative: Adaptive binning or hierarchical child markets.

11.3 Normal-Only Distributions

Problem: Only $N(\mu, \sigma)$ expressible. Alternative: Support log-normal, uniform, mixtures, or arbitrary weight vectors.

11.4 Quadratic Probability Distortion

Problem: $\hat{p}_i = x_i^2/k^2$ distorts true probabilities (Section 2.5). Alternative: Display linear probabilities $x_i/\sum x_i$ alongside quadratic market prices.

12. Comparison with Paradigm's Distribution Markets

12.1 Overview

Paradigm's paper (Dave White, Dec 2024) proposes continuous prediction markets in infinite-dimensional L^2 space. Traders hold payout functions $f: \mathbb{R} \rightarrow \mathbb{R}_+$, AMM enforces $\|f\|_2 = k$, solvency requires $\max(f) \leq b$.

12.2 Structural Comparison

Aspect	DekantPM	Paradigm
Outcome space	Finite: N bins	Infinite: $\mathbb{R}$
Invariant	$\sum x_i^2 = k^2$	$\int f^2 dx = k^2$
Solvency	Automatic	$\max(f) \leq b$
Resolution	Step function	Smooth: $f(v)$
Collateral	Complete-set mint	Max-loss based
On-chain	$O(N)$ per trade	$O(1)$ + numerics

12.3 Backing Constraint

In finite dimensions, $\|x\|_2 = k$ guarantees $|x_i| \leq k$. In infinite dimensions, $\|f\|_2 = k$ does NOT bound $\max(f)$. Hence Paradigm needs $\max(f) \leq b$ and $\sigma \geq k^2/(b^2\sqrt{\pi})$. DekantPM sidesteps this entirely.

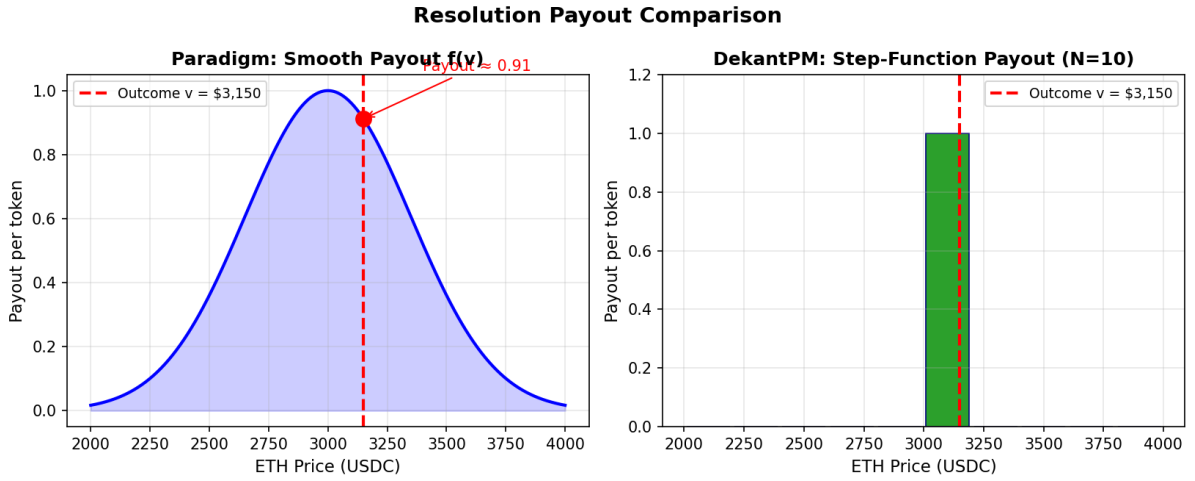


Figure 9: Paradigm gives smooth payouts; DekantPM gives winner-take-all per bin.

12.4 Convergence

As $N \rightarrow \infty$, DekantPM converges to Paradigm: sums $\rightarrow$ integrals, step payouts $\rightarrow$ smooth payouts.

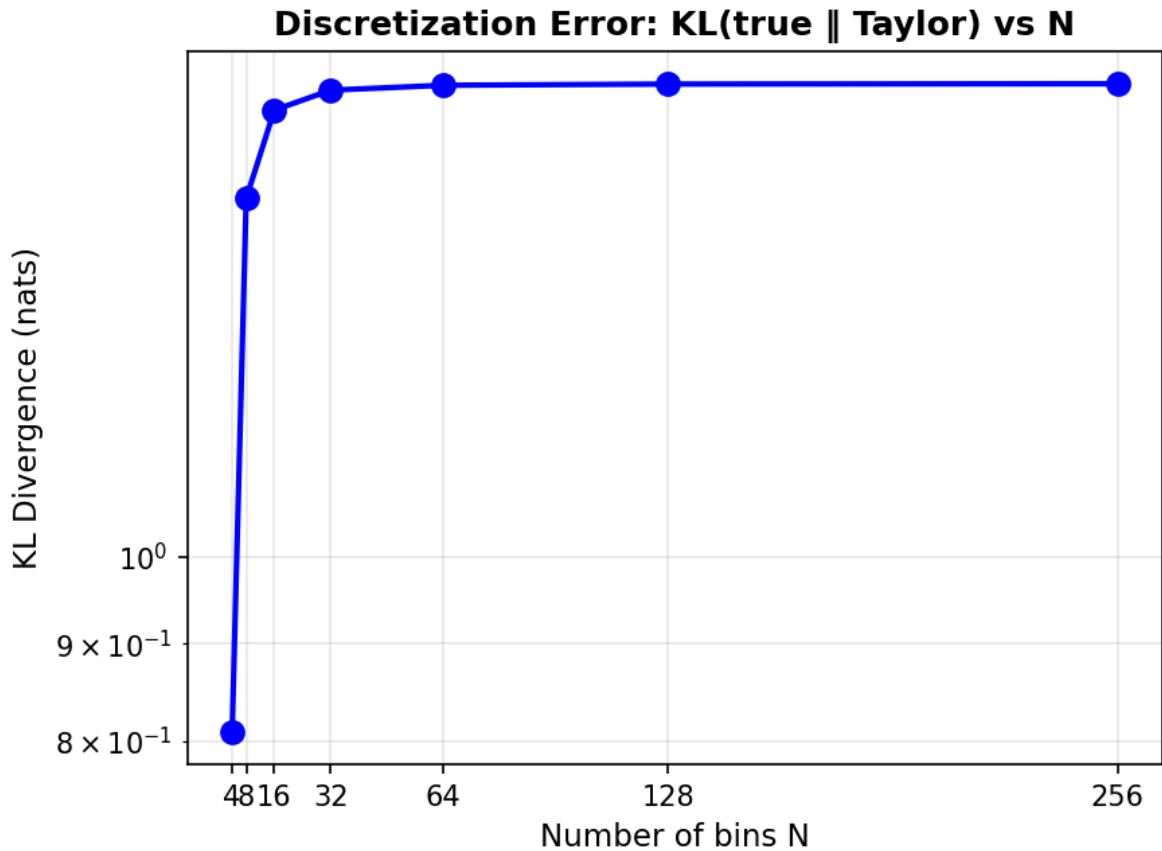


Figure 10: KL divergence between true and Taylor-approximated weights decreases with N .

13. Summary of Mathematical Properties

1. Invariant preservation: $\sum x_i^2 = k^2$ within $\varepsilon = 256$.
2. Solvency: $\text{vault} \geq k$, $\text{max redemption} \leq k$. Always solvent.
3. No arbitrage (within fees): probabilities sum to 1.
4. Market scoring rule: equilibrium $x \propto p$.

Constant	Value	Meaning
SCALE	10^9	Fixed-point denominator
INVARIANT_TOL	256	Max invariant drift
MAX_BINS	256	Max continuous bins
Z_CUTOFF	5	Gaussian tail cutoff
TRADE_FEE	30 bps	0.3% per trade
LP_FEE_SHARE	5000 bps	50% of fee to LPs
REDEMPTION_FEE	50 bps	0.5% on claims

A. Appendix: Distribution Buy Quadratic

Seek $x'_j = x_j + \lambda \cdot W_j / W^2$ satisfying $\sum (x'_j)^2 = k'^2$. Expanding and solving:

$$\lambda^2 + 2 \cdot XW \cdot \lambda - W^2 \cdot \text{excess} = 0$$

$$\lambda = \sqrt{(XW^2 + W^2 \cdot \text{excess})} - XW$$

B. Appendix: Integer Square Root

Newton's method: $x_0 = n/2 + 1$, $x_{t+1} = \lfloor (x_t + n/x_t)/2 \rfloor$. Stop when $x_{t+1} \geq x_t$. Converges in ≤ 64 iterations for u128.

C. Appendix: Glossary

Symbol	Meaning
N	Number of outcomes or bins
k	Total minted collateral
h_i, x_i	Reserve, position ($x_i = k - h_i$)
$\hat{p}_i$	Reported probability (x_i^2/k^2)
p_i	True probability ($x_i/\sum x_i$)
S	Scale factor (10^9)
W_j	Bin weight ($\sum W_j = S$)
μ, σ	Mean, std dev (trader's Normal)
XW, W^2	$\sum x_j W_j, \sum W_j^2$
λ	Distribution buy parameter
L	LP shares total